INFO282: Knowledge Representation and Reasoning (2026h)

Mandatory assignment #1 *
Logical foundations

Notes.

- Total of points: **289**.
- Deadline: 2026-09-22 (14hrs).

1 Deductive reasoning (42pts)

Exercise 1 (Valid inference and the truth-value of the involved sentences (18pts)) In each case below, assume that all involved inferences are valid. With the indicated additional assumptions, what can we say about the truth-value of the rest of the sentences? Argue *briefly* (one line for each sentence) but appropriately.

$\langle A \rangle$ **(10pts)** Suppose that A is true but G is false.

$$\cfrac{\cfrac{A \quad , \quad \cfrac{A \,,\, B \,,\, C}{D}}{E} \quad , \quad \cfrac{E}{F} \quad , \quad \cfrac{A}{B}}{G}$$

What can we say about B, C, D, E and F?

$\langle B \rangle$ **(8pts)** Suppose that A is true but F is false.

$$\cfrac{\cfrac{\cfrac{B \,,\, E}{C}}{D} \quad , \quad \cfrac{\cfrac{D}{C}}{E} \quad , \quad \cfrac{A}{B}}{F}$$

What can we say about B, C, D and E? ♦

Exercise 2 (Valid inference (24pts)) In all cases below, remember: while $P_1, \ldots, P_n \models C$ indicates that the involved inference is valid, $P_1, \ldots, P_n \not\models C$ indicates that the involved inference is NOT valid.

$\langle A \rangle$ **(8pts)** Take any sentences P, C. Suppose $P \models C$, and consider the inference $C\,/\,P$. Choose the correct option.

- ○ Under the given assumptions, $C\,/\,P$ is valid.
- ○ Under the given assumptions, $C\,/\,P$ is NOT valid.
- ○ The assumptions are not enough to guarantee that $C\,/\,P$ is valid, and also not enough to guarantee that $C\,/\,P$ is NOT valid.

$\langle B \rangle$ **(8pts)** Take any sentences P, C, Q. Suppose $P \models C$ and $C \models Q$, and consider the inference $P\,/\,Q$. Choose the correct option.

- Under the given assumptions, $P\,/\,Q$ is valid.
- Under the given assumptions, $P\,/\,Q$ is NOT valid.
- The assumptions are not enough to guarantee that $P\,/\,Q$ is valid, and also not enough to guarantee that $P\,/\,Q$ is NOT valid.

$\langle C \rangle$ **(8pts)** Take any sentences P, C, Q. Suppose $P \models C$ and $Q \not\models C$, and consider the inference $Q\,/\,P$. Choose the correct option.

- Under the given assumptions, $Q\,/\,P$ is valid.
- Under the given assumptions, $Q\,/\,P$ is NOT valid.
- The assumptions are not enough to guarantee that $Q\,/\,P$ is valid, and also not enough to guarantee that $Q\,/\,P$ is NOT valid.

In each case, justify your answer appropriately. ♦

*`INFO282-exercises.man-01-a.tex`, *compiled 19th August 2026, 10:36.*

2 Propositional logic (45pts)

Exercise 3 (Syntactic ambiguity in the propositional language (10pts)) Explain in your own words what syntactic ambiguity is. Then provide two formulas that are syntactically ambiguous, indicating why they are so. ♦

Exercise 4 (Language $\mathcal{L}_{\rightarrow,\perp}$ (20pts)) Besides connectives, one can also define *logical constants*: formulas whose truth-value is always the same. One example of this is the formula "$\perp$", which is always false.

Consider a propositional language that uses only one Boolean operator, implication ("$\rightarrow$"), and one logical constant, "$\perp$". Show that this language, denoted as $\mathcal{L}_{\rightarrow,\perp}$, is still *as expressive as* the language that uses the five Boolean operators discussed in the lecture ("$\neg$", "$\wedge$", "$\vee$", "$\rightarrow$", "$\leftrightarrow$") by providing formulas that use only "p", "q", "$\rightarrow$" and "$\perp$" (parenthesis are also allowed) and are, respectively, logically equivalent to the following ones.

$\langle A \rangle$ **(5pts)** $\neg p$ $\langle B \rangle$ **(5pts)** $p \wedge q$ $\langle C \rangle$ **(5pts)** $p \vee q$ $\langle D \rangle$ **(5pts)** $p \leftrightarrow q$

Note then what this exercise tells us: as long as we have implication ("$\rightarrow$") and the always false formula "$\perp$", we need neither negation ("$\neg$") nor conjunction ("$\wedge$") nor disjunction ("$\vee$") nor bi-implication ("$\leftrightarrow$"). ♦

Exercise 5 (Exclusive disjunction (15pts)) The propositional language discussed in the lectures uses the *inclusive* disjunction ($\vee$). According to its truth-table, $\varphi \vee \psi$ is true if and only if *at least* one of its components is true. Yet, in natural language there is also a *exclusive* disjunction (use "$\veebar$" as its symbol), which is true if and only if *exactly* one of its components is true. The truth-table below shows the difference.

φ	$\vee$	ψ	φ	$\veebar$	ψ
1	**1**	1	1	**0**	1
1	**1**	0	1	**1**	0
0	**1**	1	0	**1**	1
0	**0**	0	0	**0**	0

$\langle A \rangle$ **(5pts)** Provide a formula that uses only φ, ψ and (some of the) operators in $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$ (parenthesis are also allowed) and is logically equivalent to $\varphi \veebar \psi$. Note what this result says: as long as we have the used Boolean connectives, we do not need the exclusive disjunction.

$\langle B \rangle$ **(10pts)** Provide tableaux rules for this connective, using the kind of tree-like diagrams discussed for the standard Boolean connectives. In other words, provide a diagram indicating what to do for making $\varphi \veebar \psi$ true and a diagram indicating what to do for making said formula false. ♦

3 First-order predicate logic (80pts)

Exercise 6 (From natural language to the predicate language (30pts)) Assuming that the domain contains only persons, write the following sentences as formulas of the predicate language. Be sure to indicate what each constant symbol and each predicate label represent.

$\langle A \rangle$ **(5pts)** *"Everybody loves somebody"*.

$\langle B \rangle$ **(5pts)** *"Some boy doesn't love all girls"*.

$\langle C \rangle$ **(5pts)** *"Every boy who loves a girl is also loved by some girl"*.

$\langle D \rangle$ **(5pts)** *"Every girl who loves all boys does not love every girl"*.

$\langle E \rangle$ **(5pts)** *"No girl who does not love a boy loves a girl who loves a boy"*.

$\langle F \rangle$ **(5pts)** *"Some boy loves all girls who love only girls"*. ♦

Exercise 7 (From the predicate language to natural language (30pts)) Write the following formulas of the predicate language as sentences in natural language. Assume that the domain of discourse is the set of all persons, that **k** stands for Karen, that **j** stands for John, that **p** stands for Peter, that **SmarterThan** (x, y) is read as *"x is smarter than y"* and that **Loves** (x, y) is read as *"x loves y"*.

$\langle A \rangle$ **(5pts)** $\exists x\,.\,\forall y\,.\,(\mathbf{Loves}(x,y) \rightarrow \mathbf{Loves}(y,x))$

$\langle B \rangle$ **(5pts)** $\forall x\,.\,(\mathbf{Girl}(x) \rightarrow \exists y\,.\,\mathbf{Loves}(y,x))$

$\langle C \rangle$ **(5pts)** $\neg\exists x\,.\,(\mathrm{Girl}\,(x) \wedge \forall y\,.\,\neg\,\mathrm{Loves}\,(x,y))$

$\langle D \rangle$ **(5pts)** $\forall x\,.\,(\exists y\,.\,\mathrm{Loves}\,(x,y) \rightarrow \exists z\,.\,\mathrm{Loves}\,(z,x))$

$\langle E \rangle$ **(5pts)** $\exists y\,.\,\forall x\,.\,\mathrm{Loves}\,(y,x)$

$\langle F \rangle$ **(5pts)** $\forall y_1\,.\,((\mathrm{Girl}\,(y_1) \wedge \exists y_2\,.\,(\mathrm{Girl}\,(y_2) \wedge \mathrm{Loves}\,(y_1,y_2))) \rightarrow \exists x\,.\,(\mathrm{Boy}\,(x) \wedge \mathrm{Loves}\,(x,y_1)))$ ♦

Exercise 8 (Building models (20pts)) Consider the following predicate formulas. For each one of them, build a model in which the given formula is true but the other three are all false. If you think you cannot build one of such models, justify appropriately.

$\langle A \rangle$ **(5pts)** $\forall x\,.\,\exists y\,.\,\mathrm{R}\,(x,y)$ $\langle B \rangle$ **(5pts)** $\forall x\,.\,\exists y\,.\,\mathrm{R}\,(y,x)$ $\langle C \rangle$ **(5pts)** $\exists x\,.\,\forall y\,.\,\mathrm{R}\,(x,y)$ $\langle D \rangle$ **(5pts)** $\exists x\,.\,\forall y\,.\,\mathrm{R}\,(y,x)$ ♦

4 Deciding logical consequence (70pts)

Exercise 9 (Valid inference (16pts))

$\langle A \rangle$ **(8pts)** Consider the inference $\varphi \wedge \neg\varphi\,/\,\psi$. Choose the correct option.

- ○ $\varphi \wedge \neg\varphi\,/\,\psi$ is valid for any φ, ψ.
- ○ $\varphi \wedge \neg\varphi\,/\,\psi$ is NOT valid for every φ, ψ.
- ○ $\varphi \wedge \neg\varphi\,/\,\psi$ is valid for some φ, ψ, and NOT valid for some φ, ψ.

$\langle B \rangle$ **(8pts)** Consider the inference $\varphi\,/\,\psi \vee \neg\psi$. Choose the correct option.

- ○ $\varphi\,/\,\psi \vee \neg\psi$ is valid for any φ, ψ.
- ○ $\varphi\,/\,\psi \vee \neg\psi$ is NOT valid for every φ, ψ.
- ○ $\varphi\,/\,\psi \vee \neg\psi$ is valid for some φ, ψ, and NOT valid for some φ, ψ.

In each case, justify your answer appropriately. ♦

Exercise 10 (Logical consequence (24pts)) Consider the following four propositional formulas.

- $p \wedge (q \rightarrow r)$
- $(p \wedge q) \rightarrow r$
- $p \rightarrow (q \wedge r)$
- $p \rightarrow (q \rightarrow r)$

Between *every pair* of formulas $\langle \varphi_1, \varphi_2 \rangle$ add, in following diagram, an arrow "$\rightarrow$" if and only if $\varphi_1 \models \varphi_2$ (that is, if and only if φ_2 is a logical consequence of φ_1).

$\boxed{(p \wedge q) \rightarrow r}$

$\boxed{p \wedge (q \rightarrow r)}$ $\boxed{p \rightarrow (q \rightarrow r)}$

$\boxed{p \rightarrow (q \wedge r)}$ ♦

Exercise 11 (Logical consequence (30pts)) Decide whether each one of the following inferences are valid. In each case: if you answer is *"yes"*, justify it (e.g., show the tree generated by the tableaux method, or provide a brief but convincing argument), and if your answer is *"no"*, provide a counterexample.

$\langle A \rangle$ **(5pts)** $\forall x\,.\,(\mathrm{P}\,(x) \vee \mathrm{Q}\,(x))\,/\,\forall x\,.\,\mathrm{P}\,(x) \vee \exists x\,.\,\mathrm{Q}\,(x)$

$\langle B \rangle$ **(5pts)** $\exists x\,.\,(\mathrm{P}\,(x) \wedge \mathrm{Q}\,(x))\,/\,\exists x\,.\,\mathrm{P}\,(x) \wedge \exists x\,.\,\mathrm{Q}\,(x)$

$\langle C \rangle$ **(5pts)** $\exists x\,.\,\mathrm{P}\,(x) \wedge \exists x\,.\,\mathrm{Q}\,(x)\,/\,\exists x\,.\,(\mathrm{P}\,(x) \wedge \mathrm{Q}\,(x))$

$\langle D \rangle$ **(5pts)** $\exists x\,.\,\mathrm{P}\,(x) \wedge \forall x\,.\,\mathrm{Q}\,(x)\,/\,\exists x\,.\,(\mathrm{P}\,(x) \wedge \mathrm{Q}\,(x))$

$\langle E \rangle$ **(5pts)** $\forall x\,.\,\forall y\,.\,(\mathrm{R}\,(x,y) \rightarrow \neg\,\mathrm{R}\,(y,x))\,/\,\forall x\,.\,\neg\,\mathrm{R}\,(x,x)$

$\langle F \rangle$ **(5pts)** $\forall x\,.\,\mathrm{P}\,(x) \rightarrow \forall x\,.\,\mathrm{Q}\,(x)\,/\,\forall x\,.\,(\mathrm{P}\,(x) \rightarrow \mathrm{Q}\,(x))$ ♦

5 Default reasoning (52pts)

Exercise 12 (Comparing $\models$ and $\approx_{cwa}$ (24pts)) This exercise asks you to compare the deduction consequence relation "$\models$" with the default *cwa* consequence relation "$\mathrel{|}\!\approx_{cwa}$". Let Φ be finite set of propositional formulas; let ψ be a propositional formula.

$\langle A \rangle$ **(12pts)** Suppose that $\Phi \models \psi$. Is this enough to guarantee $\Phi \mathrel{|}\!\approx_{cwa} \psi$? If your answer is *"yes"*, argue convincingly; if your answer is *"no"*, provide a counterexample (a finite set of formulas Φ and a formula ψ such that $\Phi \models \psi$ and yet $\Phi \not\mathrel{|}\!\approx_{cwa} \psi$).

$\langle B \rangle$ **(12pts)** Suppose that $\Phi \mathrel{|}\!\approx_{cwa} \psi$. Is this enough to guarantee $\Phi \models \psi$? If your answer is *"yes"*, argue convincingly; if your answer is *"no"*, provide a counterexample (a finite set of formulas Φ and a formula ψ such that $\Phi \mathrel{|}\!\approx_{cwa} \psi$ and yet $\Phi \not\models \psi$). ♦

Exercise 13 (Comparing $\models$ and $\mathrel{|}\!\approx_{\leqslant}$ (28pts)) This exercise asks you to compare the deduction consequence relation "$\models$" with the default plausibility consequence relation "$\mathrel{|}\!\approx_{\leqslant}$". Let Φ be finite set of propositional formulas; let ψ be a propositional formula.

$\langle A \rangle$ **(7pts)** Suppose that $\Phi \models \psi$. Is this enough to guarantee $\Phi \mathrel{|}\!\approx_{\leqslant} \psi$ for *every* plausibility ordering $\leqslant$? If your answer is *"yes"*, argue convincingly; if your answer is *"no"*, provide a counterexample (a finite set of formulas Φ and a formula ψ such that $\Phi \models \psi$ and yet $\Phi \not\mathrel{|}\!\approx_{\leqslant} \psi$).

$\langle B \rangle$ **(7pts)** Suppose that $\Phi \mathrel{|}\!\approx_{\leqslant} \psi$ holds for *every* plausibility ordering $\leqslant$. Is this enough to guarantee $\Phi \models \psi$? If your answer is *"yes"*, argue convincingly; if your answer is *"no"*, provide a counterexample (a finite set of formulas Φ and a formula ψ such that $\Phi \mathrel{|}\!\approx_{\leqslant} \psi$ for every plausibility ordering $\leqslant$, and yet $\Phi \not\models \psi$).

$\langle C \rangle$ **(7pts)** Suppose that $\Phi \models \psi$. Is this enough to guarantee $\Phi \mathrel{|}\!\approx_{\leqslant} \psi$ for *some* plausibility ordering $\leqslant$? If your answer is *"yes"*, argue convincingly; if your answer is *"no"*, show that, under the given conditions, under any plausibility order you can find a valuation makes all formulas in Φ true but ψ false.

$\langle D \rangle$ **(7pts)** Suppose that $\Phi \mathrel{|}\!\approx_{\leqslant} \psi$ holds for *some* plausibility ordering $\leqslant$. Is this enough to guarantee $\Phi \models \psi$? If your answer is *"yes"*, argue convincingly; if your answer is *"no"*, provide a counterexample (a finite set of formulas Φ, a formula ψ *and a plausibility ordering* $\leqslant$ such that $\Phi \mathrel{|}\!\approx_{\leqslant} \psi$ and yet $\Phi \not\models \psi$). ♦